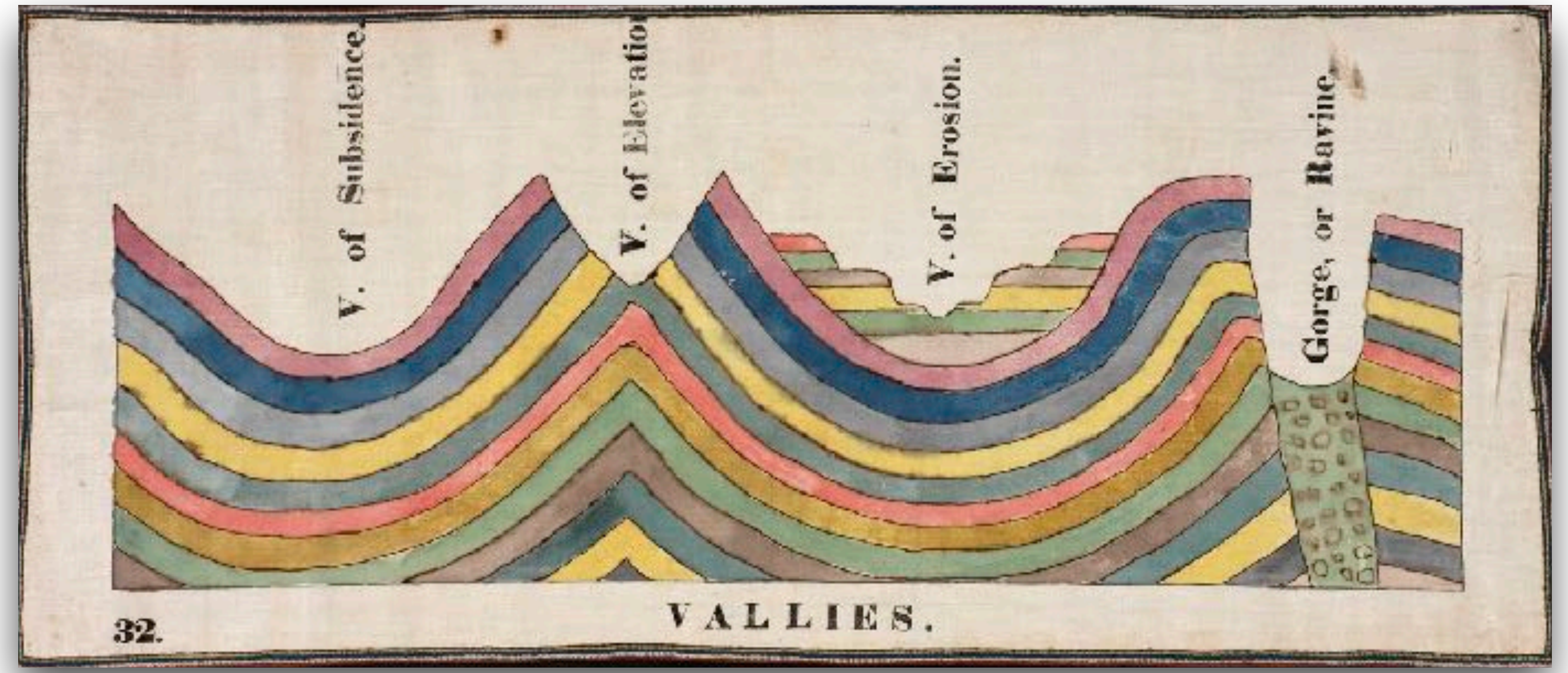
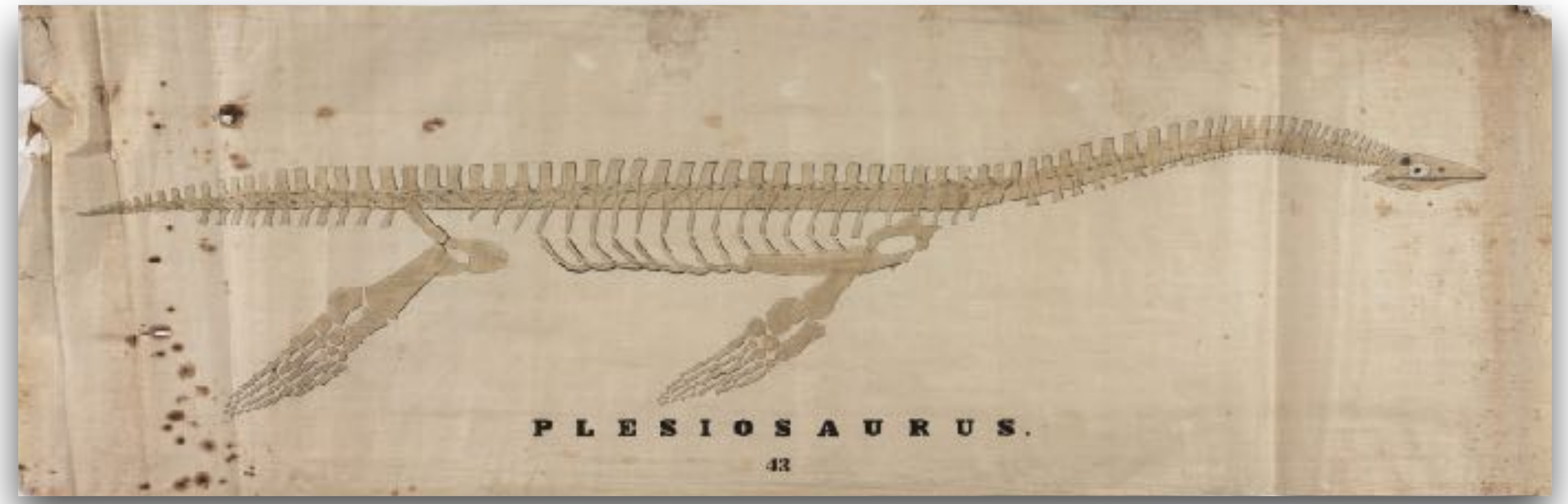
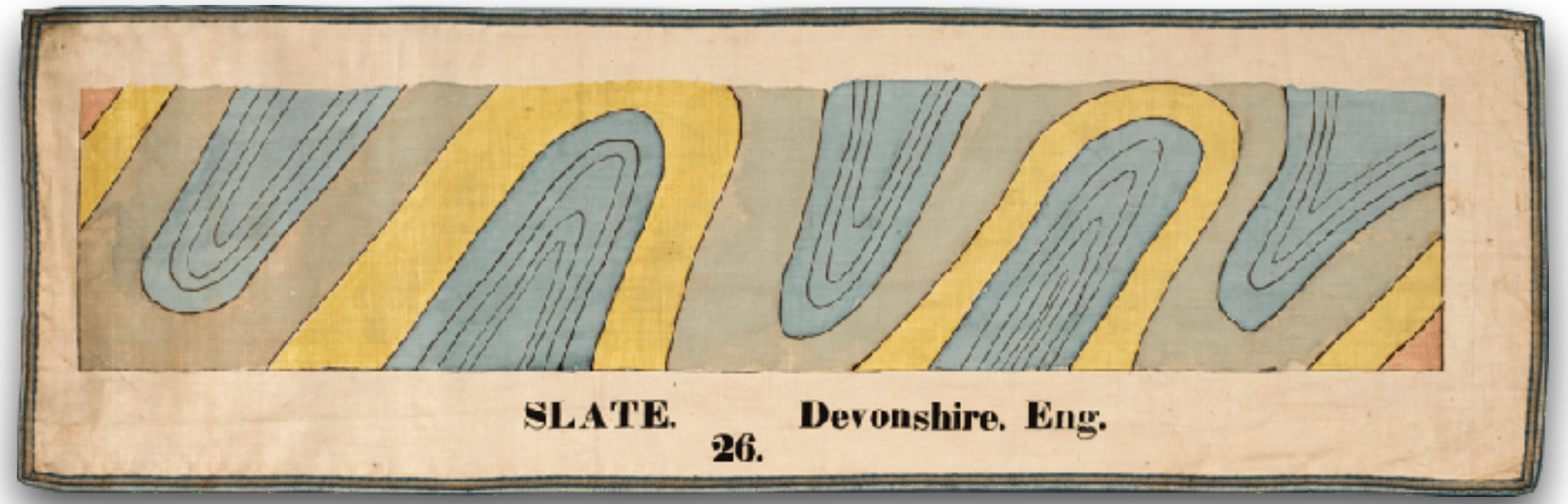






modelling trait evolution on phylogenies

There are no better words than those of **Luke Harmon** to kickstart this:

Comparative methods combine biology, mathematics, and computer science to learn about a wide variety of topics in evolution using phylogenetic trees and other associated data.

For example, we can find out which processes must have been common, and which rare, across clades in the tree of life …

and whether the evolutionary potential that we see playing out in real time is sufficient to explain the diversity of life on earth, or whether we might need additional processes that may come into play only very rarely or over very long timescales.

What can you do with phylogenetic comparative methods?

- model fitting - testing evolutionary hypotheses on how traits evolve.
- ancestral state(s) reconstructions - **will do in the practical!**
- diversification - speciation and extinction rates.
- biogeography 🌎.
- …

Character types:

- **Discrete characters:** traits with a finite number of distinct states. *Examples:* presence or absence of wings, flower color categories.

- **Continuous characters:** traits that vary along a measurable scale. *Examples:* body length, enzyme activity levels, beak depth.

Analytical approaches:

- **Discrete characters:** analyzed using models like the Mk model, which describes the evolution of discrete traits as a Markov process.

- **Continuous characters:** often examined with models such as Brownian Motion (BM) and Ornstein-Uhlenbeck (OU) processes.

Discrete characters

The most commonly used model for discrete character evolution on trees is a model called the **Mk model**.

- **M** for the process modelled follows the Markov property
- **k** as the model is generalized to include an arbitrary number (k) states

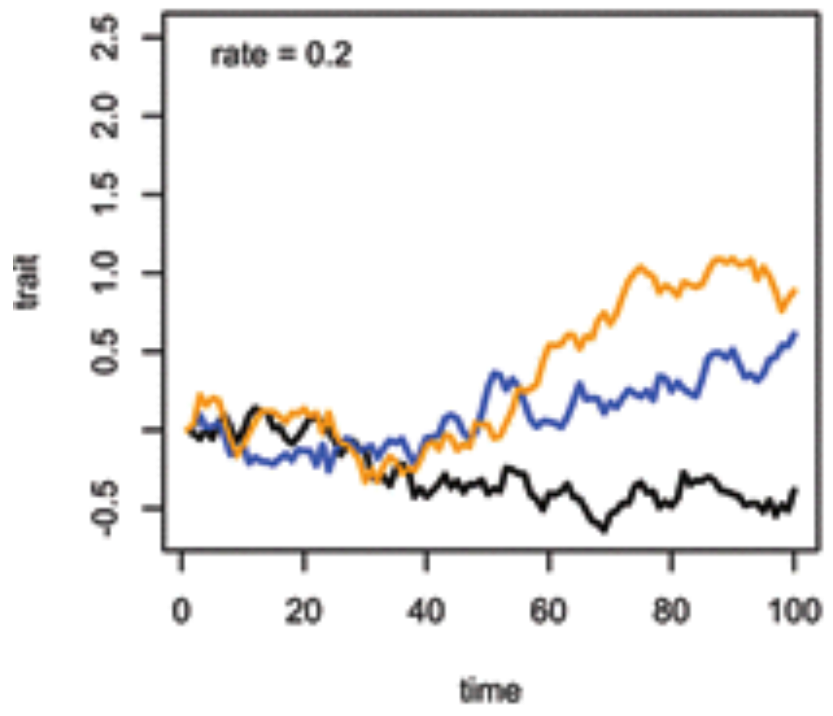
… **does it remind you anything ?!**

Depending on the assumptions we make about the transition rates, the Mk model can be parameterised in several different ways:

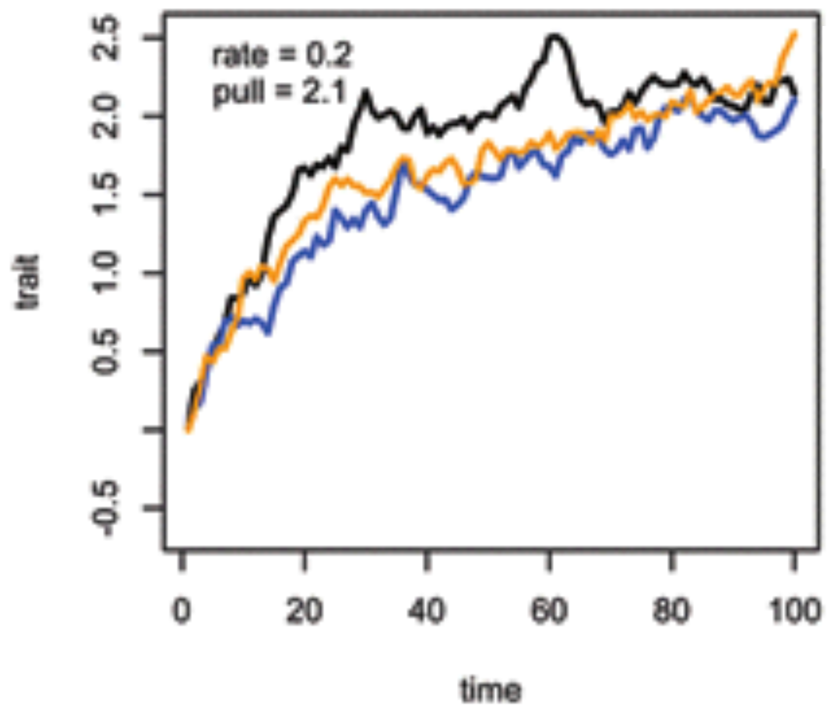
- **ER** - Equal Rates: all transitions have the same rate.
- **SYM** - Symmetrical: different rates but forward and reverse trans. equal.
- **ARD** - All Rates Different: each transition has its own rate.

Models of continuous trait evolution

- **Brownian Motion (BM)**:
  trait evolve randomly and variance increases with time



- **Ornstein-Uhlenbeck (OU)**:
  add stabilizing selection and traits evolve toward an optimum



Models of continuous trait evolution

What is Brownian Motion (BM)?

- a random walk model of trait evolution.
- at each step, trait changes by a small and normal random amount.
- no preferred direction or optimum - purely stochastic.

Key features

- trait variance increases linearly with time since divergence.
- closely related species tend to have more similar traits.
- trait evolution is neutral: no selection toward an optimum.

When to use BM?

- BM is a **null model** for trait evolution.
- It is a good fit for traits evolving neutrally or with drift.

Models of continuous trait evolution

What is the OU model?

- An extension of Brownian Motion that includes stabilizing selection.
- Traits evolve with a tendency toward an optimal value θ.

Key features

- Variance increases with time but levels off as values are “pulled” toward the optimum.
- Includes three parameters:
  - **θ**: trait optimum
  - **z_0**: the ancestral/root state
  - **α**: strength of selection toward the optimum
  - **σ2**: evolutionary rate - as in BM!

When to use OU?

- When traits show selective constraints or adaptive peaks.
- For testing **adaptive hypotheses** or detecting **convergent evolution**.

What is phylogenetic signal?

- The tendency for related species to resemble each other more than expected under a model of random trait distribution.
- Reflects the **non-independence of species due to shared ancestry.**

Why it matters?

- **Strong signal**: trait variation is structured by phylogeny.
- **Weak signal**: trait may evolve rapidly or independently of phylogeny.
- Helps choose the right model and correct for autocorrelation.

Pagel's λ (Lambda)

- A branch-length scaling parameter that quantifies how much the trait variation is explained by the phylogeny.
- Modifies the internal branches of the tree to best match the observed trait covariance.

How it works

The internal branches of a phylogenetic tree represent the shared evolutionary history between species.

Pagel's λ rescales these internal branches to reflect how much shared ancestry influences trait similarity. Bounded between 0 & 1.

$\lambda = 1$ - full phylogenetic covariance - trait evolves under Brownian motion.
$\lambda = 0$ - internal branches collapsed - trait evolves independently of phylo.

Blomberg's K

- A variance-based metric that quantifies how much trait similarity among species is explained by their shared evolutionary history.

- Specifically, it compares observed trait variance across a phylogeny to the variance expected under Brownian motion (BM) evolution.

How it works

- The core concept: $K = \dfrac{\text{Observed phylogenetic signal}}{\text{Expected signal under Brownian motion}}$

- $K \approx 1$ Trait matches BM expectations
- $K > 1$ Trait is more conserved than expected
- $K < 1$ Trait is less conserved (weak signal or convergence)

a phylogenetic problem: non-independence in data!

Species are not independent data points! 🐍🦎🧸🦧🦍

They share evolutionary history! 🐟🐡🐠🦀🪲

… and this creates statistical non-independence.

What happens if we disregard phylogeny?

- Violates the assumption of independently distributed residuals.
- inflated Type I error rates - false positives!

Use PIC and PGLS!

Phylogenetically Independent Contrasts (PIC)

- Developed by Joe Felsenstein (1985)
- Data transformation + regression
- Traits evolve under Brownian motion
- Simple, effective, fast

Phylogenetic Generalized Least Squares (PGLS)

- Developed by Grafen (1989)
- Models covariance structure directly in regression
- Generalized least squares regression
- Flexible: can include λ, OU models

FINISH